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Chapter 1: Introduction

1.1 Project Summary

The 3D Royal Mansion Interactive Portfolio is an immersive, single-page web application designed to transform traditional, static portfolio websites into an interactive room-scale 3D experience. Built using HTML5, WebGL, Three.js, and GSAP, the application renders a 3D royal hall featuring detailed architectural elements, dynamic lighting, interactive chamber doors, profile art frames, and custom wall sconces. Users navigate through the space using familiar keyboard controls (WASD / Arrow Keys) and mouse interactions to explore portfolio categories such as About Me, Projects, Skills, Awards, Documents, and Contact.

1.2 Problem Statement

Traditional 2D web portfolios often rely on flat text, standard cards, and repetitive web layouts. As a result, hiring managers and recruiters may find them less engaging, and static layouts do not effectively demonstrate advanced WebGL, 3D graphics, scene design, and interactive development skills. Users are also mainly passive readers rather than active explorers. The proposed system addresses these limitations through an immersive 3D portfolio environment.

1.3 Objectives

1. Build an immersive 3D royal hall with high-quality textures and realistic lighting.
2. Enable intuitive navigation using WASD / Arrow movement and mouse camera control.
3. Provide interactive chamber doors for portfolio categories.
4. Implement smooth camera and door animations using GSAP.
5. Generate dynamic canvas textures for door plates and profile wall art.
6. Maintain high performance and compatibility across modern WebGL-capable browsers.

1.4 Scope

In Scope:

1. Client-side WebGL rendering using Three.js.
2. Interactive camera movement, boundary control, and raycasting hover tooltips.
3. Wall sconce lighting with glowing torch meshes and point lights.
4. Camera interpolation and door opening animations using GSAP.
5. Dynamic HTML5 Canvas textures for door plates and portrait frames.
6. Responsive viewport resizing and input handling.
7. Portfolio sections: About Me, Projects, Skills, Awards, Documents, and Contact.

Out of Scope:

1. Multiplayer or networking capabilities.
2. Backend database storage.
3. Native desktop application packaging.

Chapter 2: System Analysis

2.1 Existing System Analysis

Current personal portfolios are predominantly built as flat 2D web pages using standard HTML/CSS layouts or page builders. They are generally quick to load, easy to edit, and simple to navigate. However, they often provide limited interaction, repetitive layouts, and fewer opportunities to demonstrate real-time 3D rendering capabilities.

2.2 Proposed System

The proposed 3D Royal Mansion Interactive Portfolio replaces standard scrolling pages with a spatial 3D gallery environment. Users can explore a virtual mansion hall, interact with chamber doors, view dynamic tooltips, and experience smooth camera and door animations. Different chambers represent different portfolio categories.

2.3 Feasibility Analysis

Technical Feasibility:

Modern web browsers support WebGL, while Three.js and GSAP provide mature tools for 3D rendering and animation.

Operational Feasibility:

The system uses familiar WASD/Arrow controls, mouse interaction, visual tooltips, and on-screen instructions.

Economic Feasibility:

The project uses freely available web technologies and open-source libraries, reducing licensing costs.

Schedule Feasibility:

The project can be developed iteratively by building the scene, lighting, navigation, interactions, animations, and optimization in stages.

2.4 Requirement Analysis

The system must efficiently handle 3D rendering, user input, camera movement, raycasting, lighting, textures, animations, and portfolio information within a browser-based environment. Requirements were defined according to the expected experience of visitors and the portfolio owner.

2.5 System Requirements

2.5.1 Functional Requirements

1. 3D Scene Initialization
2. Floor Texture
3. Ceiling and Fog
4. Room Lighting
5. Wall Sconces
6. Spotlight
7. Keyboard Navigation
8. Camera Control
9. Boundary Control
10. Interactive Doors
11. Hover Feedback
12. Raycasting Interaction
13. Door Animation
14. Door Labels
15. Profile Artwork
16. Portfolio Section Access
17. Window Resize

2.5.2 Non-Functional Requirements

1. Performance
2. Loading Performance
3. Usability
4. Readability
5. Browser Compatibility
6. Responsiveness
7. Reliability
8. Input Handling

2.5.3 Hardware and Software Requirements

Category	Requirement
Processor	Dual-Core 2.0 GHz or higher
RAM	Minimum 4 GB; recommended 8 GB+
Graphics	WebGL-supported integrated or dedicated GPU
Operating System	Windows, macOS, Linux, Android, or iOS
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari
Rendering	WebGL

3D Library	Three.js r128
Camera Control	OrbitControls
Animation	GSAP 3.x
Typography	Google Fonts
Icons	FontAwesome 6

2.6 System Methodology

2.6.1 SDLC Model

The project follows an Agile / Iterative Development Model. The system is developed in small iterations, allowing continuous improvement of scene design, lighting, interaction, performance, and user experience.

1. Requirements and architecture setup
2. 3D scene design and lighting
3. Navigation and interaction development
4. GSAP door and camera animation
5. Testing, performance optimization, and refinement

2.6.2 Requirement Gathering

Requirements were gathered through analysis of modern interactive portfolio concepts, 3D web design practices, expected visitor navigation, and performance considerations for desktop and mobile WebGL-capable browsers.

2.6.3 Key Users and Stakeholders

User / Stakeholder	Role
Visitor	Explores the 3D mansion and views portfolio information.
Administrator	Manages portfolio information, project details, profile information, and chamber labels.
Recruiter / Client	Reviews portfolio, skills, projects, achievements, and contact information.

2.6.4 SRS Table

Functional Requirements

FR001	3D Scene Initialization
Description	The system shall create a 40×40 unit royal room with floor, ceiling, and four walls.
Actor	Visitor, Administrator
FR002	Room Lighting
Description	The system shall provide ambient, fill, and spotlight lighting for the royal hall.
Actor	Visitor
FR003	Wall Sconces
Description	The system shall provide twin-torch wall sconces with warm point lights.
Actor	Visitor
FR004	Keyboard Navigation
Description	The system shall allow users to move using W, A, S, D and Arrow Keys.
Actor	Visitor
FR005	Camera Control
Description	The system shall allow users to rotate the camera using mouse interactions.
Actor	Visitor
FR006	Boundary Control
Description	The system shall restrict camera movement within the defined room boundaries.
Actor	Visitor
FR007	Interactive Doors
Description	The system shall provide interactive chamber doors for different portfolio sections.
Actor	Visitor
FR008	Hover Feedback
Description	The system shall display tooltips and pointer feedback when the user hovers over interactive objects.
Actor	Visitor
FR009	Raycasting Interaction
Description	The system shall detect user interaction with interactive 3D objects using raycasting.
Actor	Visitor
FR010	Door Animation

Description	The system shall animate the selected door and camera when the user clicks a door.
Actor	Visitor

FR011	Door Labels
Description	The system shall display room numbers and chamber titles on the doors.
Actor	Visitor, Administrator

FR012	Profile Artwork
Description	The system shall display the profile image, name, and title on the royal wall artwork.
Actor	Visitor, Administrator

FR013	Portfolio Section Access
Description	The system shall provide access to About Me, Projects, Skills, Awards, Documents, and Contact sections.
Actor	Visitor

FR014	Window Resize
Description	The system shall adjust the camera and renderer when the browser window is resized.
Actor	Visitor

FR015	Manage Portfolio Information
Description	The administrator shall be able to update portfolio information, project details, and chamber labels.
Actor	Administrator

Non-Functional Requirements

NFR001	Performance
Description	The system should maintain approximately 50–60 FPS on standard consumer devices.
Actor	Visitor

NFR002	Loading Performance
Description	The system shall use optimized textures to reduce loading time and memory usage.
Actor	Visitor

NFR003	Usability
Description	The system shall provide clear navigation instructions and easy-to-understand interactions.
Actor	Visitor

NFR004	Readability
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Description	The system shall maintain readable labels, tooltips, and interactive elements.
Actor	Visitor

NFR005	Browser Compatibility
Description	The system shall support modern WebGL-capable web browsers.
Actor	Visitor

NFR006	Responsiveness
Description	The system shall adjust the renderer and camera according to the browser window size.
Actor	Visitor

NFR007	Reliability
Description	The system shall operate correctly during normal user interaction without unexpected failure.
Actor	Visitor, Administrator

NFR008	Input Handling
Description	The system shall ignore navigation key presses when the user is typing in an input field or text area.
Actor	Visitor

Chapter 3: References

3.1 Reference

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